DAA LAB ASSIGNMENT 2

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Section-c

Semester -5th (3rd year)

Que-1 write a program for insertion sort

```
#include<stdio.h>
int main(){
 int n,a[10],i,j,key;
 printf("enter the size of array");
 scanf("%d",&n);
 printf("enter the elements of array ");
 for(i=0;i<n;i++){
   scanf("%d",&a[i]);
 }
 for(j=1;j<n;j++){
   key=a[j];
   i=j-1;
   while(i>=0 && a[i]>key){
      a[i+1]=a[i];
```

```
      i=i-1;
    }
    a[i+1]=key;
  }
  printf("print array after sorting");
  for(i=0;i<n;i++){
    printf("%d ",a[i]);
  }
  return 0;
}
```

```
enter the size of array 6
enter the elements of array 34 234 57 599 45 233
print array after sorting34 45 57 233 234 599
PS C:\Users\91843\Desktop\DSA LAB>
```

Que -2 write a program for selection sort

```
#include <stdio.h>
void selection(int arr[], int n)
{
    int i, j, small;

    for (i = 0; i < n-1; i++)
```

```
    {
        small = i;

        for (j = i+1; j < n; j++)
        if (arr[j] < arr[small])
            small = j;

    int temp = arr[small];
    arr[small] = arr[i];
    arr[i] = temp;
    }
}

void printArr(int a[], int n)
{
    int i;
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
}
```

```
int main()
{
   int a[] = { 12, 31, 25, 8, 32, 17 };
   int n = sizeof(a) / sizeof(a[0]);
   printf("Before sorting array elements are - \n");
   printArr(a, n);
   selection(a, n);
   printf("\nAfter sorting array elements are - \n");
   printArr(a, n);
   return 0;
}
```

```
Before sorting array elements are -
12 31 25 8 32 17
After sorting array elements are -
8 12 17 25 31 32
PS C:\Users\91843\Desktop\DSA LAB>
```

Que -3 Write a program to implement quick sort

```
#include<stdio.h>
void swap(int* a, int* b) {
   int t = *a;
```

```
    *a = *b;
    *b = t;
}
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = (low - 1);

    for (int j = low; j <= high - 1; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
```

```
      quickSort(arr, pi + 1, high);
   }
}
void printArray(int arr[], int size) {
   int i;
   for (i = 0; i < size; i++)
      printf("%d ", arr[i]);
   printf("\n");
}

int main() {
   int arr[] = { 12, 17, 6, 25, 1, 5 };
   int n = sizeof(arr) / sizeof(arr[0]);
   quickSort(arr, 0, n - 1);
   printf("Sorted array: \n");
   printArray(arr, n);
   return 0;
}
```